SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

COMPUTER SCIENCE AND ENGINEERING PROGRAMME

CSA3717 – SOFTWARE TESTING

EXPERIMENT 15

COMPREHENSIVE TEST PLAN – INVENTORY MANAGEMENT SYSTEM

Aim:

To Develop a comprehensive Test Plan document for an Inventory Management System, covering test scope, test design techniques, test execution strategy, and test completion criteria.

1. Introduction

The Inventory Management System maintains product details, stock levels, supplier information, purchase orders, goods receipt, stock issue, stock adjustment and inventory reports. This plan describes how to validate the correctness and reliability of inventory operations.

2. Test Objectives

- Verify that the system satisfies documented functional and non-functional requirements.
- Identify defects before release and provide reproducible evidence.
- Check data accuracy, access control, usability and reliability.
- Confirm that fixes do not introduce regressions.

3. Test Scope

In-scope activities:

- Product and category management
- Supplier management
- Purchase order creation and approval
- Goods receipt and stock updates
- Stock issue, transfer and adjustment
- Low-stock alerts and reorder levels
- Inventory valuation and stock reports
- User roles and audit history

Out of scope: production data modification, live financial transactions, third-party services that are not available in the test environment, and features not included in the approved requirements.

4. Features to be Tested

Module / Feature	Testing Focus
Product Master	SKU uniqueness, mandatory fields, category and unit validation.
Supplier Management	Supplier details, status and contact validation.
Purchase Orders	Order creation, approval, quantity and price calculations.
Goods Receipt	Received quantity, partial receipt and stock update.
Stock Issue / Transfer	Available quantity checks and location-wise updates.
Alerts and Reports	Reorder threshold alerts, stock summary and movement reports.

5. Test Strategy and Methodology

Testing Level / Type	Planned Activities
Unit Testing	Developers test individual functions, validations and calculations.
Integration Testing	Verify communication between modules, databases and connected services.
System Testing	Execute end-to-end workflows against the complete application.
Regression Testing	Repeat selected tests after defect fixes or code changes.
Usability Testing	Check clarity of navigation, labels, messages and accessibility.
Security Testing	Check authentication, authorization, session handling and input validation.
Performance Testing	Measure response time and stability under expected concurrent usage.

6. Test Design Techniques

Technique	Application
Equivalence Partitioning	Divide inputs into valid and invalid groups and test representative values.
Boundary Value Analysis	Test minimum, maximum and values immediately below/above limits.
Decision Table Testing	Test combinations of conditions that determine an outcome.
State Transition Testing	Verify valid and invalid transitions between workflow states.
Use-Case Testing	Validate complete user journeys from start to completion.

7. Test Environment and Test Data

- Use a dedicated staging environment that is separate from production.
- Use synthetic accounts, records and transactions; do not expose real customer data.
- Record browser/device, operating system, application build and database version.
- Prepare valid, invalid, boundary and role-specific test data.
- Enable application logs and test evidence capture.

8. Test Execution Strategy

1. Map each requirement to one or more test conditions.
2. Create test products, suppliers, warehouses and opening stock quantities.
3. Test normal purchase, receipt, issue, transfer and adjustment workflows.
4. Apply boundary tests around zero stock, reorder level and maximum quantity.
5. Test concurrent stock updates and verify that totals remain consistent.
6. Retest fixes, run regression tests and prepare the final test summary.

9. Entry and Exit Criteria

Entry Criteria	**Exit / Completion Criteria**
Approved requirements, stable test build, configured test warehouse, user roles and synthetic product/supplier data are available.	All critical inventory workflows pass; stock balances reconcile across transactions; no unresolved critical defects remain; reports and audit records are verified; test evidence is archived.

10. Defect Management

Each defect must include a unique ID, title, environment, preconditions, reproducible steps, expected result, actual result, severity, priority, evidence and current status. Defects move through New → Assigned → In Progress → Fixed → Retesting → Closed. Failed retests are reopened.

11. Risks and Mitigation

Risk	Mitigation
Stock mismatch	Reconcile opening stock and transaction ledger after every test.
Concurrent update conflict	Test simultaneous receipts/issues and verify atomic stock updates.
Incorrect reorder alert	Test values below, equal to and above reorder threshold.
Duplicate SKU	Enforce uniqueness and test duplicate product creation.
Unauthorized stock adjustment	Verify role permissions and audit every adjustment.

12. Deliverables

- Approved test plan and requirement coverage list.
- Test scenarios, test cases and test data.
- Execution log with Pass / Fail / Blocked status.
- Defect reports with supporting evidence.
- Test summary report and release recommendation.

13. Conclusion

This test plan defines a structured approach for evaluating the Inventory Management System. Applying the planned test levels, design techniques, execution controls and completion criteria helps improve product quality and reduces the risk of defects reaching users.